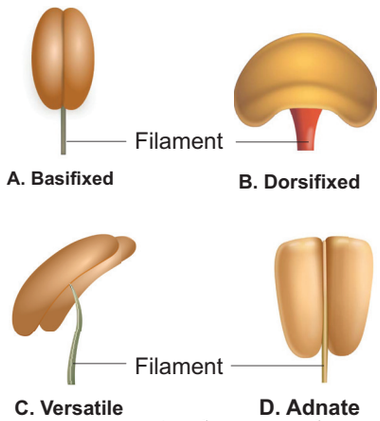


Figure 4.20: Anther attachment

4.5 Gynoecium

Gynoecium or pistil is the female reproductive part of the flower.

A pistil consists of an expanded basal portion called the ovary, an elongated section called a **style** and an apical structure that receives pollen called a **stigma**. Ovary with stipe is called **stipitate ovary.**

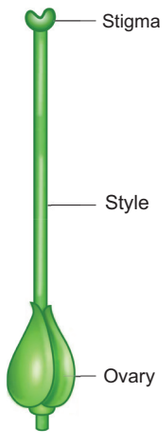


Figure 4.21: Pistil

Carpel: They are components of a gynoecium. Gynoecium is made of one or more carpels. Carpels may be distinct or connate.

4.5.1 Number of carpel

Unicarpellary (monocarpellary) Single carpel Example: Fabaceae	**Bicarpellary** Two carpels Example: Rubiaceae
Tricarpellary Three carpels Example: Cucurbitaceae	**Tetracarpellary** Four carpels Example: Lamiaceae.
Multicarpellary Many carpels Example: Nymphaeceae.	

4.5.2 Fusion of carpels

It is an important systematic character. Apocarpous gynoecium is generally thought to be ancestral condition in Angiosperms.

Apocarpous	**Syncarpous**
A pistil contains two or more distinct carpels. Example: *Annona*.	A pistil contains two or more carpels which are connate. Example: *Citrus*, Tomato.

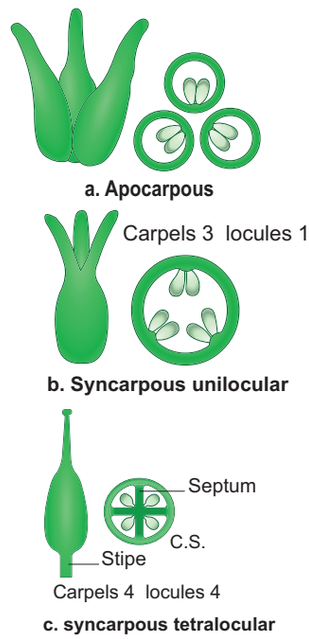


Figure 4.22: Fusion of carpels

4.5.3 Number of locules

Ovary bears ovules on a specialized tissue called **placenta**. A **septum** is a crosswall or partition of ovary. The walls of ovary and septa form a cavity called **locule**. Like that tetralocular and pentalocular ovaries are present according to the locule numbers four or five. More than one locule ovaries are called **plurilocular**.

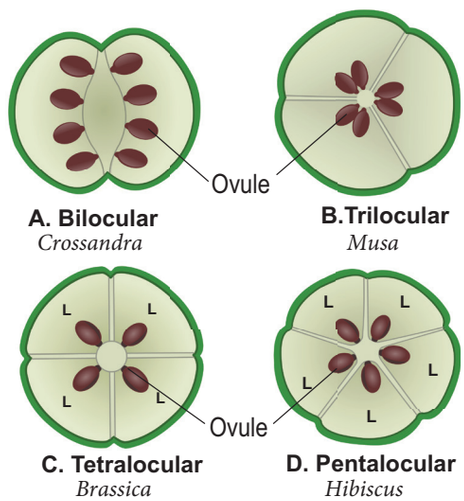


Figure 4.23: Locules

11th Botany_EM_Chapter 4.indd 89 05-01-2023 18:24:05